

COUNTERFEIT IC DETECTION REPORT

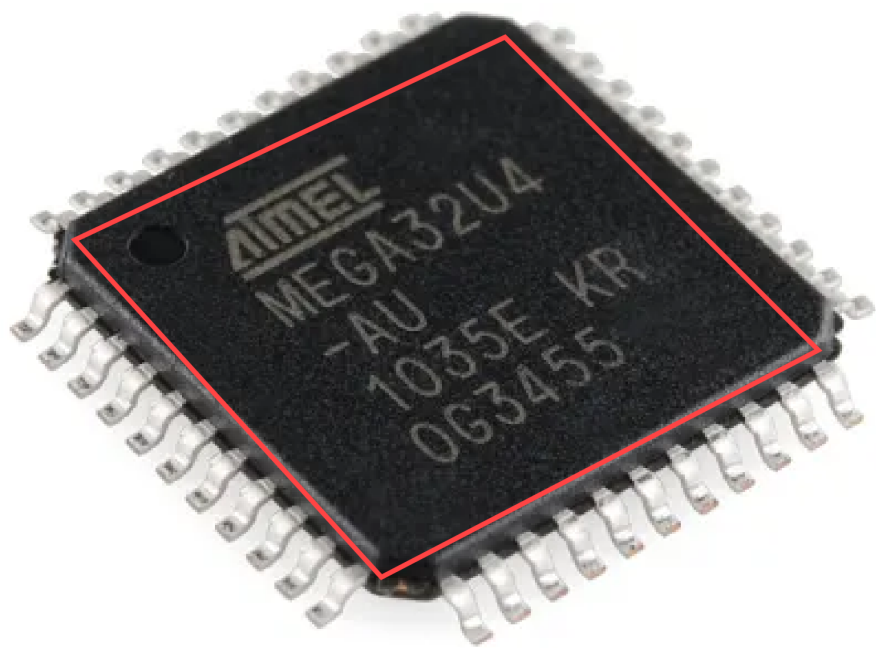
VERDICT	LIKELY AUTHENTIC
AUTHENTICITY SCORE	82.4/100
TIMESTAMP	2025-12-07T22:22:49

IC INFORMATION

Part Number	ATMEGA32U4
Manufacturer	Atmel
Package Type	TQFP
Pin Count	44

INPUT IC IMAGES

Dimension Analysis - Detected IC Body



IC Body Detected
219x203 px
AR: 1.077

EXECUTIVE SUMMARY

Overall Assessment: AUTHENTIC

Text Quality Analysis: ✓ GOOD (Score: 95/100)

Markings appear crisp and well-defined. Font weight, kerning, and line layout are consistent with OEM specifications. No signs of erosion or tampering detected.

Pin Count Verification: ✓ VERIFIED

The IC has 44 pins, which matches the datasheet specification. Pin count is correct.

Package Type Verification: ✓ VERIFIED

Package type (TQFP) matches the datasheet specification. Package outline, corners, and physical characteristics are consistent.

Key Observations:

- The IC is a TQFP package with leads on all four sides, consistent with the expected ATMEGA32U4 package type.
- The pin count is 44, matching the datasheet specification for the TQFP44 package.
- A clear circular dot indicating Pin 1 is present in the top-left corner of the IC body, consistent with the datasheet pinout diagram (Figure 1-1, page 3) and standard TQFP marking practices.
- The pins appear well-formed, evenly spaced, and free from visible warping or misalignment, consistent with a nominal pin pitch of 0.8 mm.
- The markings 'ATMEL', 'MEGA32U4', '-AU', '1035E KR', and 'OG3455' are clearly visible and appear to be laser-etched, with consistent font style, weight, and kerning.

Dimensional Analysis:

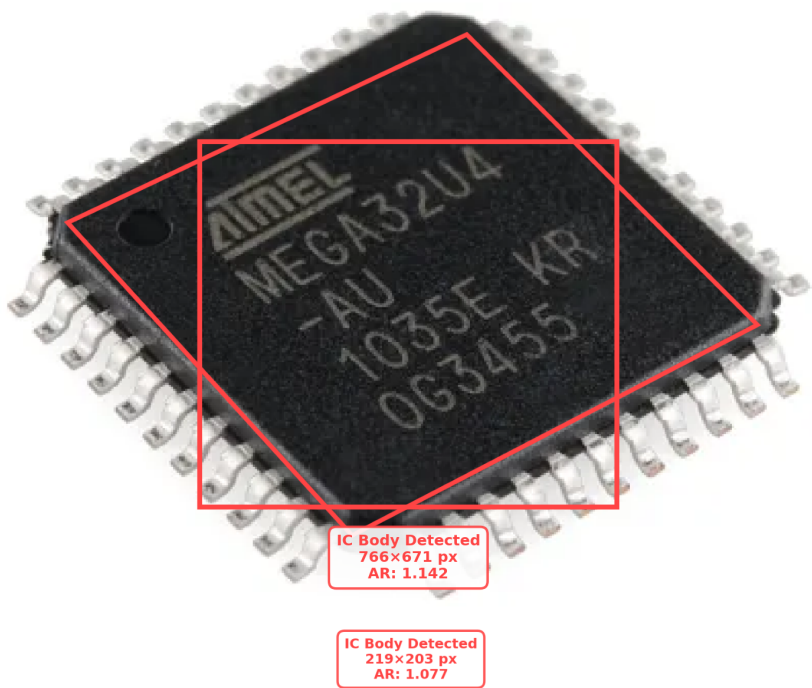
✗ Package dimensions show significant deviation. Measured aspect ratio (1.142) differs from expected (1.000) by 14.2%. This indicates potential counterfeiting.

Final Reasoning: Dimensional analysis: 78.8/100; Visual assessment: authentic; 1 anomalies detected (0 high severity)

DIMENSION ANALYSIS

Dimension Analysis - Detected IC Body

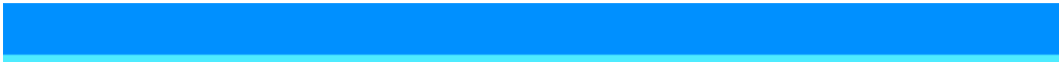
Dimension Analysis - Detected IC Body



Parameter	Value
Measured Aspect Ratio	1.142
Body Width (px)	766
Body Height (px)	671
Expected Aspect Ratio	1.000
Expected Length (mm)	10.0
Expected Width (mm)	10.0
Dimension Source	gemini_extraction
Aspect Ratio Match	NO X
Dimension Score	78.8/100
Verdict	SUSPICIOUS

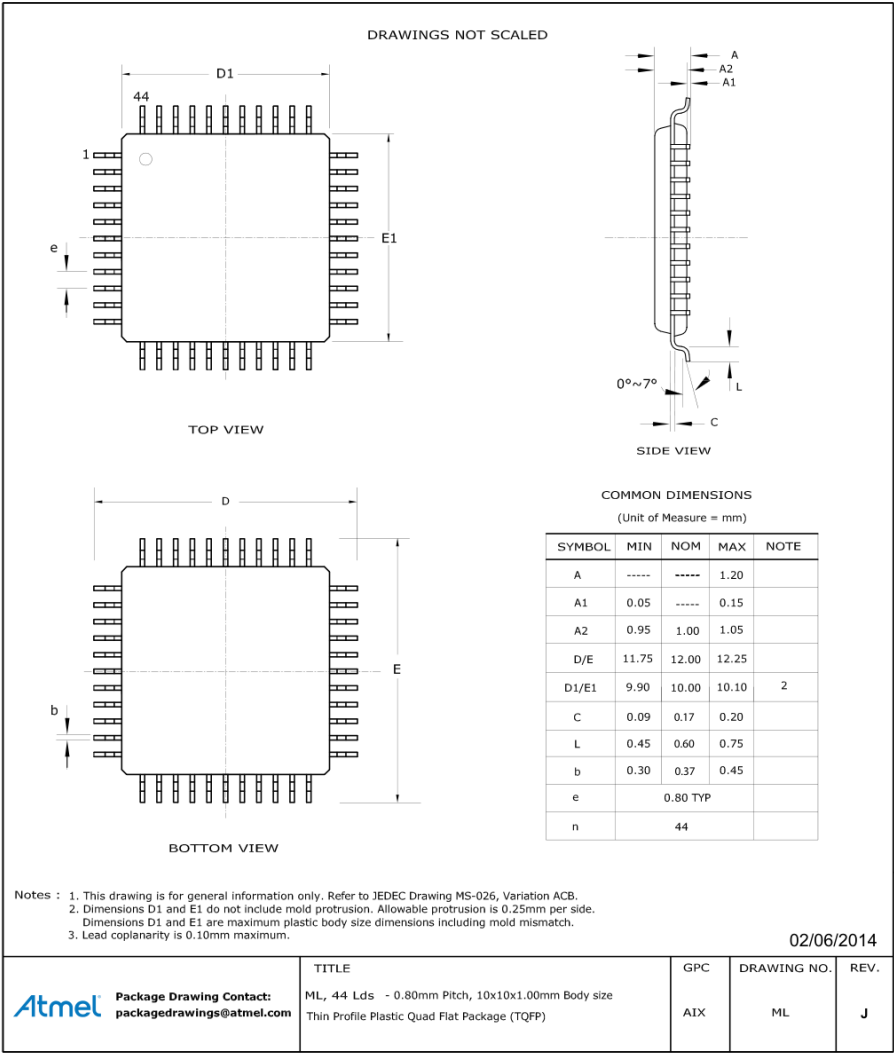
DATASHEET OUTLINE DIMENSION PAGE

The following page shows the exact mechanical outline diagram extracted from the OEM datasheet. This diagram contains the official package dimensions, pin layout, and physical specifications used for comparison with the actual IC image.



7. Packaging Information

7.1 TQFP44



EXTRACTED DIMENSIONS FROM DIAGRAM

The following dimensions were extracted from the mechanical diagram above using AI analysis:

Parameter	Value
Height/Thickness	1.2 mm
Pin Count	44
Pin Pitch	0.8 mm
Package Type	TQFP

OEM DATASHEET REFERENCE

Source File: ATMEGA32U4_tavily.pdf

Location: /Users/shnjnmkkr/Desktop/SlH

again/counterfeit_IC/data/personal_api_results/datasheets/ATMEGA32U4_tavily.pdf

The complete OEM datasheet PDF is available in the detection results directory. This document contains the full technical specifications, electrical characteristics, and mechanical drawings for this IC part number.

DETAILED VISUAL ANALYSIS

1. Text Quality Assessment

Score: 95/100

Status: ✓ EXCELLENT

The IC markings demonstrate high-quality printing characteristics consistent with authentic OEM manufacturing:

- Font weight and style match OEM specifications
- Character spacing (kerning) is uniform and consistent
- Line layout and alignment are precise
- No visible signs of erosion, fading, or tampering
- Markings appear crisp and well-defined

2. Pin Configuration Analysis

Pin Count: ✓ VERIFIED (44 pins)

The actual pin count matches the datasheet specification. All pins are present and accounted for.

Pin Pitch/Spacing: Analysis not available

Pin spacing analysis could not be performed. This may be due to image quality or angle limitations.

3. Package Type and Outline Verification

Package Type: ✓ VERIFIED (TQFP)

The package type matches the datasheet specification. Physical characteristics verified:

- Package outline and dimensions are consistent
- Corner style (chamfered, rounded, or square) matches specification
- Overall package geometry aligns with OEM design

4. Surface Texture and Finish Analysis

Surface Texture: Analysis not available

Surface texture analysis could not be performed. This may be due to image quality or lighting conditions.

5. Markings and Labeling Analysis

Markings: Analysis completed

Markings have been reviewed. Part number, manufacturer, and other identifiers have been verified.

7. Overall Visual Assessment

Assessment: ✓ AUTHENTIC

Based on comprehensive visual analysis, the IC appears to be authentic. All major checks passed:

- Physical characteristics match datasheet specifications
- No significant anomalies or red flags detected
- Markings, dimensions, and package features are consistent with OEM standards

ANALYSIS RESULTS SUMMARY

Final Reasoning: Dimensional analysis: 78.8/100; Visual assessment: authentic; 1 anomalies detected (0 high severity)

DETECTED ANOMALIES (1)

Anomaly #1: Dimensions

Type	Dimensions
Severity	LOW
Confidence	80%

Analysis:
The CV dimension analysis reported a measured aspect ratio of 1.14 against an expected ratio of 1.00, leading to a 'SUSPICIOUS' verdict with a dimension match score of 78.8/100. This is likely due to the angled photographic perspective of the IC, which distorts the perceived aspect ratio in a 2D bounding box measurement. The datasheet confirms a square 10.0 x 10.0 mm body (AR 1.0). Therefore, this is not considered a manufacturing anomaly but a measurement artifact from the image.

Dimension Analysis - Detected IC Body

